AGREEMENT OF THE TWO CONSTRUCTIONS OF DERIVED LOWER SHRIEK FOR QUASI-FINITE SEPARATED MORPHISMS

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Allowed standard prerequisites (editorial).

- Section constructing the compactification lower shriek functors (Section section-derived-lower-shriek-compactification, canonical tag 0F7H). cited at statement lines 3841-3842. One of the two constructions whose agreement is claimed.
- Section constructing derived duality for locally quasi-finite morphisms (Section section-derived-duality-locally-quasi-finite, canonical tag 0F5M). cited at statement lines 3844-3845. The other construction of f_! on derived categories; the claimed agreement is with its restriction.
- Zariski main theorem factorization through a finite morphism (More on Morphisms, Lemma more-morphisms-lemma-quasi-finite-separated-pass-through-finite, canonical tag 05K0). cited at proof lines 3850-3853. Supplies the factorization f = f-bar after j with j an open immersion and f-bar finite.
- Vanishing of higher direct images for finite morphisms (Etale Cohomology, Proposition etale-cohomology-proposition-finite-higher-direct-image-zero, canonical tag 03QP). cited at proof lines 3855-3857. Since f-bar is finite, Rf-bar_* = f-bar_*.
- Compactification description of f_! (Lemma lemma-compactify-f-shriek-separated, canonical tag 0F52). cited at proof lines 3858-3859. Gives f-bar_* after j_! = f_!, closing the agreement.

Source reference. Original section 0F7H: sources/stacks-chow/tags.txt, source lines 19007–19007. Cited at statement lines 3841-3842: one of the two constructions whose agreement is claimed.

Source reference. Original section 0F5M: sources/stacks-chow/tags.txt, source lines 18942–18942. Cited at statement lines 3844-3845: the other construction of f_! on derived categories; the claimed agreement is with its restriction.

Source reference. Original lemma 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Cited at proof lines 3850-3853: supplies the factorization f = \overline{f} after j with j an open immersion and \overline{f} finite.

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Source reference. Original proposition 03QP: sources/stacks-chow/tags.txt, source lines 4902–4902. Cited at proof lines 3855-3857: since \overline{f} is finite, R\overline{f}_* = \overline{f}_*.

Source reference. Original lemma 0F52: sources/stacks-chow/tags.txt, source lines 18922–18922. Cited at proof lines 3858-3859: gives \overline{f}_* after j_! = f_!, closing the agreement.

Lemma 0.2. *Let $f : X \to Y$ be a quasi-finite separated morphism of quasi-compact and quasi-separated schemes. Then the functors $Rf_!$ constructed in Section 0F7H agree with the restriction of the functor $f_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ constructed in Section 0F5M to their common domains of definition.*

Proof. By Zariski's main theorem (More on Morphisms, Lemma 05K0) we can find an open immersion $j : X \to \overline{X}$ and a finite morphism $\overline{f} : \overline{X} \to Y$ with $f = \overline{f} \circ j$. By construction we have $Rf_! = R\overline{f}_* \circ j_!$. Since $\overline{f}$ is finite, we have $R\overline{f}_* = \overline{f}_*$ by Étale Cohomology, Proposition 03QP. The lemma follows because $\overline{f}_* \circ j_! = f_!$ for example by Lemma 0F52. □